

THE KINKAID SCHOOL • PHYSICS I • UNIT 1: KINEMATICS • 2025–2026

PACKET 6 — Free Fall & Projectiles

NAME _____ PERIOD _____ DATE _____

6.1 Free fall in one dimension

PROJECTILES

If the **only** force acting on an object is the force of gravity, that object is a **projectile**. All projectiles experience the same vertical acceleration toward the ground:

$$a = g = 9.8 \text{ m/s}^2 \approx 10 \text{ m/s}^2$$

We will give a technical definition of forces and explain this later on. For now: any object flying through the air that does not experience significant air resistance and has no other obvious force acting on it — a rocket engine, a propeller — is a projectile.

42. A raccoon is standing next to a well. It tosses a stone upwards with a vertical velocity of 15 m/s and the stone falls into the well.

a) How long will it take for the stone to rise to its highest point?

b) How high will the stone rise?

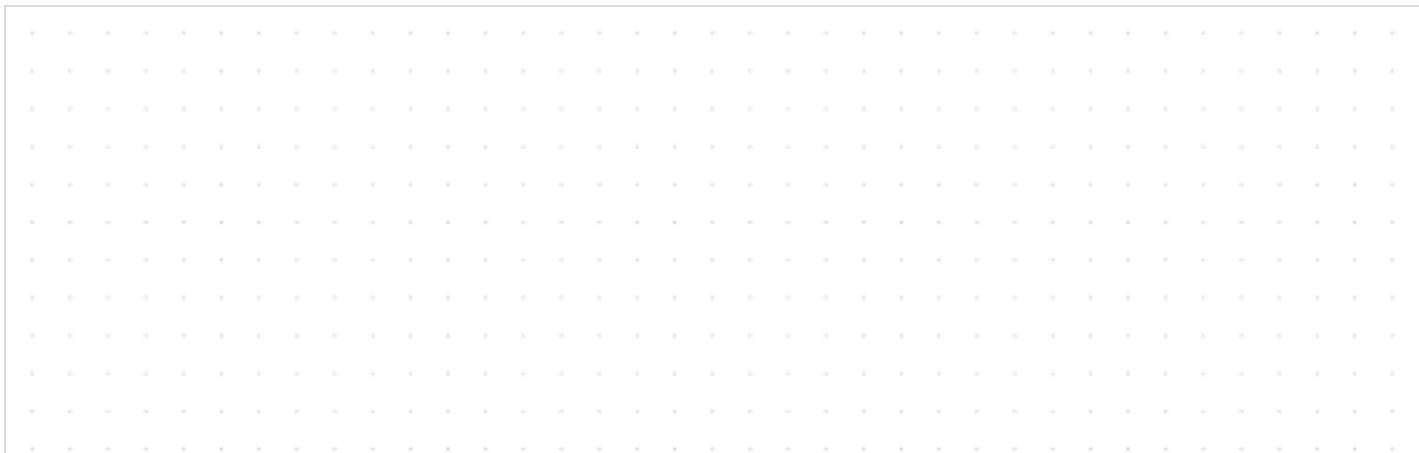
c) The well is 20 meters deep. How long after throwing the stone will the raccoon hear a splash?

d) What will the velocity of the stone be immediately before it hits the water?

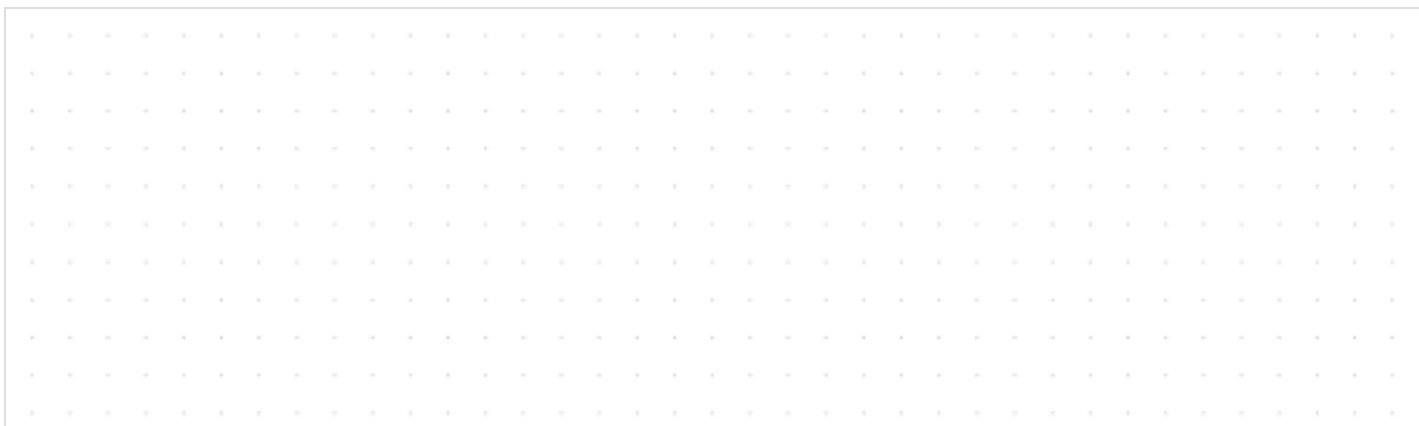
e) If instead of throwing the stone 15 m/s upwards the raccoon throws it 15 m/s *downwards*, what will the velocity of the stone be at the moment it hits the water?

43. A mountain climber stands at the top of a 50.0 m cliff hanging over a calm pool of water. The climber throws two stones vertically 1.0 s apart and observes that they cause a single splash when they hit the water. The first stone has an initial velocity of +2.0 m/s.

a) How long after the release of the first stone will the two stones hit the water?



b) What is the initial velocity of the second stone when it is thrown?



c) What will the velocity of each stone be at the instant both stones hit the water?

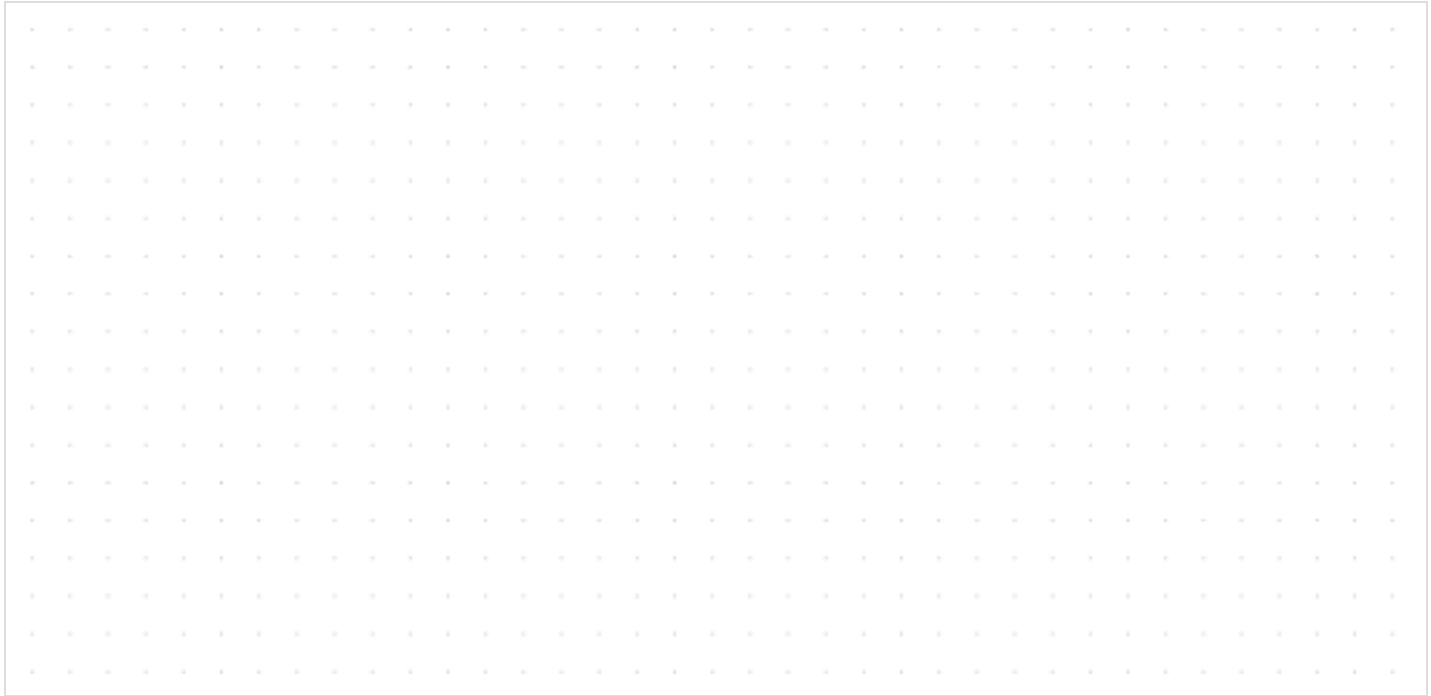


NOTES, WORK & ANNOTATIONS

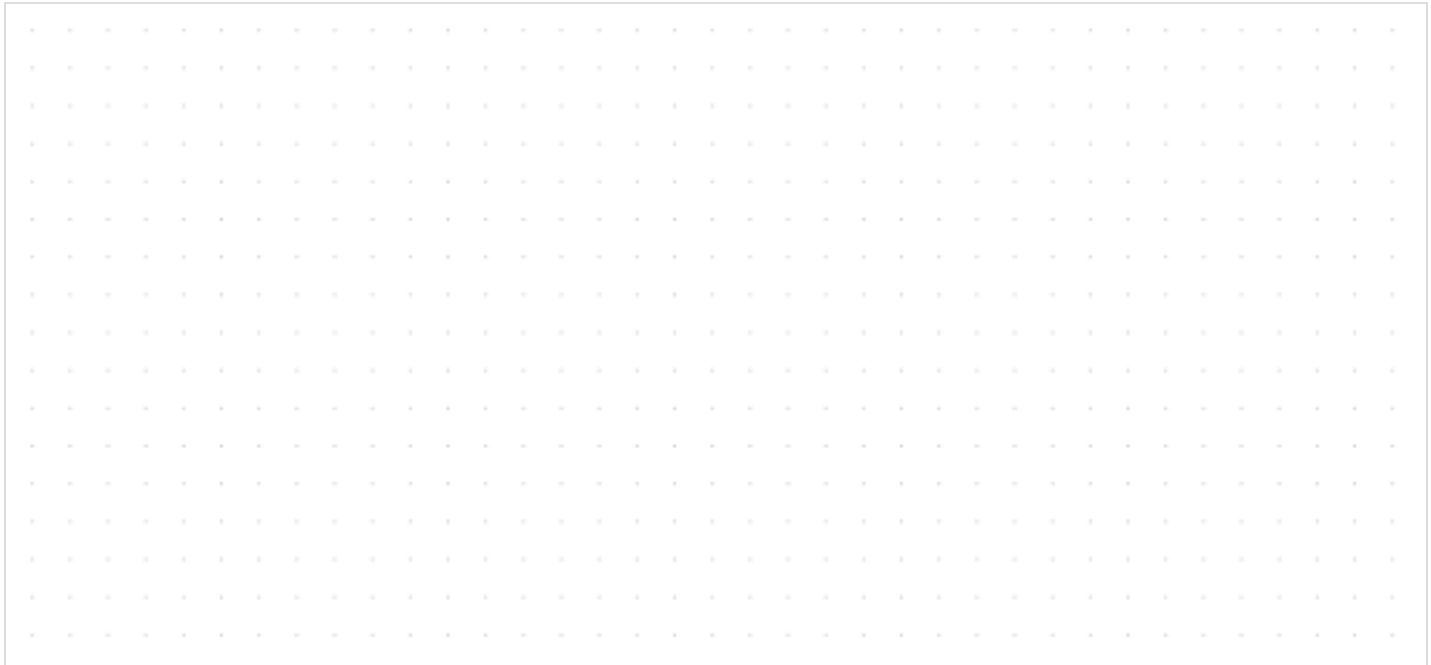


44. A model rocket is launched upwards with an initial speed of 50 m/s. It accelerates with a constant upwards acceleration of 2 m/s^2 until its engines stop at an altitude of 150 m.

a) What is the maximum height reached by the rocket?



b) How long is the rocket in the air?



This is a two-interval problem, and the acceleration is different in each interval. Before you compute anything, write down what a is while the engine burns and what a is after it stops.

6.2 Vectors in two dimensions

ADDING PERPENDICULAR DISPLACEMENTS

When two displacements are at right angles, the straight-line distance between start and finish is the **magnitude of the resultant vector**:

$$|\Delta \vec{r}| = \sqrt{\Delta x^2 + \Delta y^2}$$

and its direction is found from

$$\tan \theta = \frac{\Delta y}{\Delta x}$$

The same two relationships turn a pair of perpendicular velocity components into a speed and a launch angle — you will use them in §6.3.

45. Hans takes a walk around the big city. He walks 400 m to the West and 600 m to the North. Let East and North be positive. What is the distance between his initial position and his final position?

NOTES, WORK & ANNOTATIONS

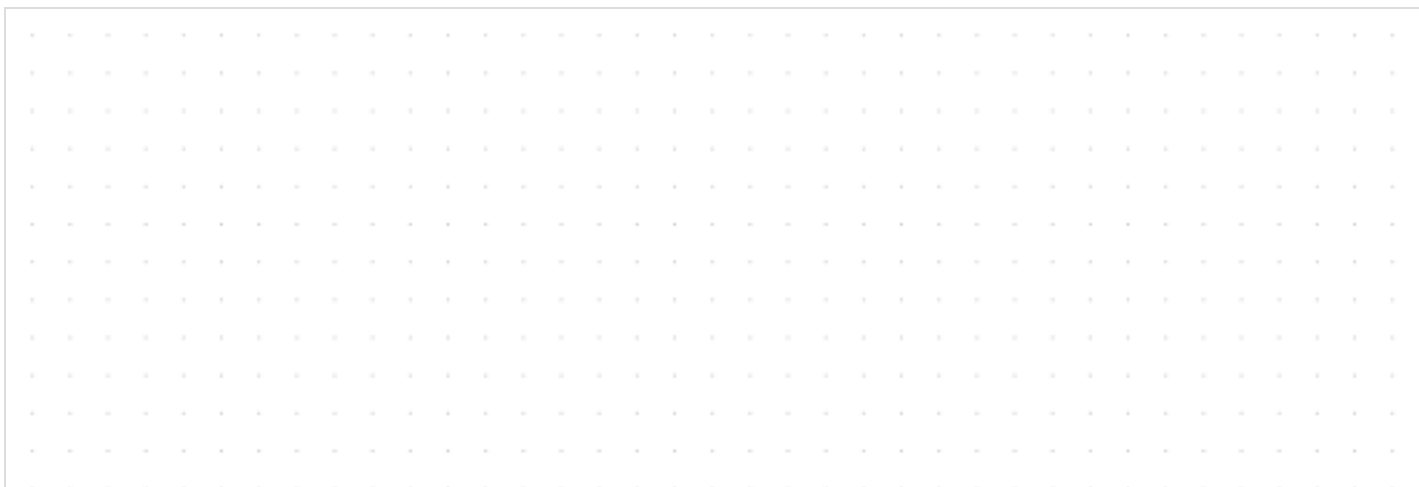
6.3 Two-dimensional projectile motion

INDEPENDENCE OF THE SPATIAL DIMENSIONS OF MOTION

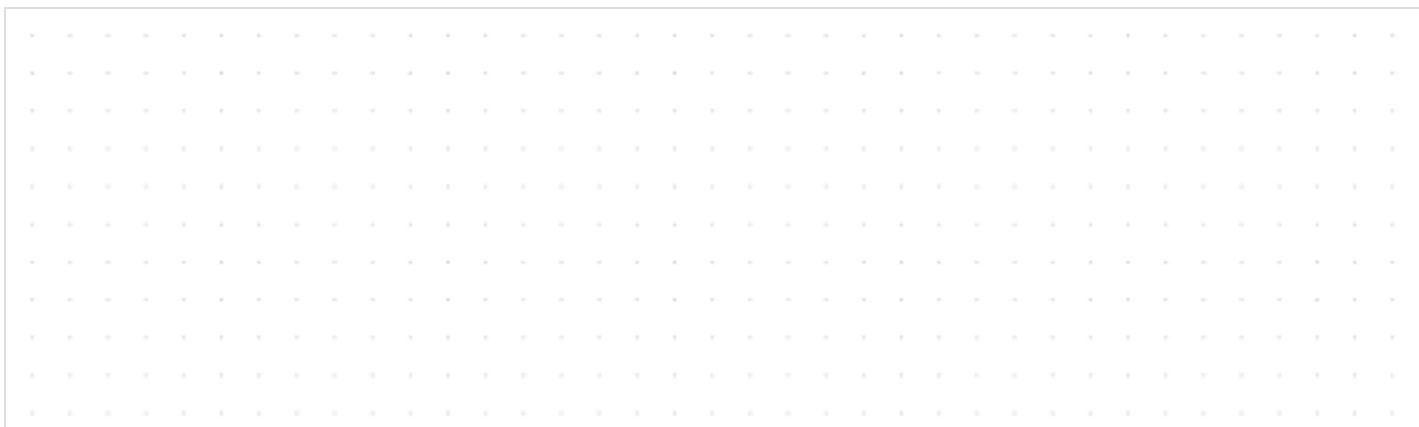
The horizontal and vertical components of two-dimensional motion are **independent**. In general, motion in the horizontal direction does not affect motion in the vertical direction, and vice versa.

The one thing the two directions share is the **clock**. Time is the bridge between them.

46. A ball is rolling on a flat cubical box at a speed of 5 m/s. The box is 2 m on each side. It reaches the edge of the box and begins to fall. How far from the base of the box will the ball land?



47. A cannonball is fired horizontally from a cannon at the edge of a cliff. The cliff is 100 m tall. The cannonball lands 300 m away from the base of the cliff. What was the initial velocity of the cannonball?

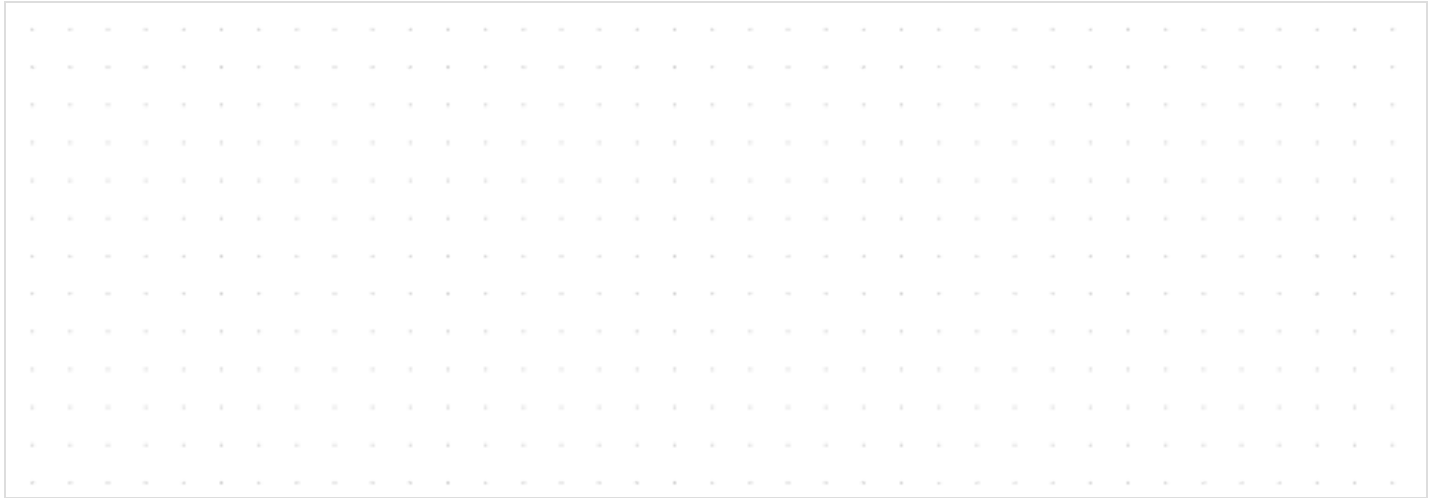


NOTES, WORK & ANNOTATIONS

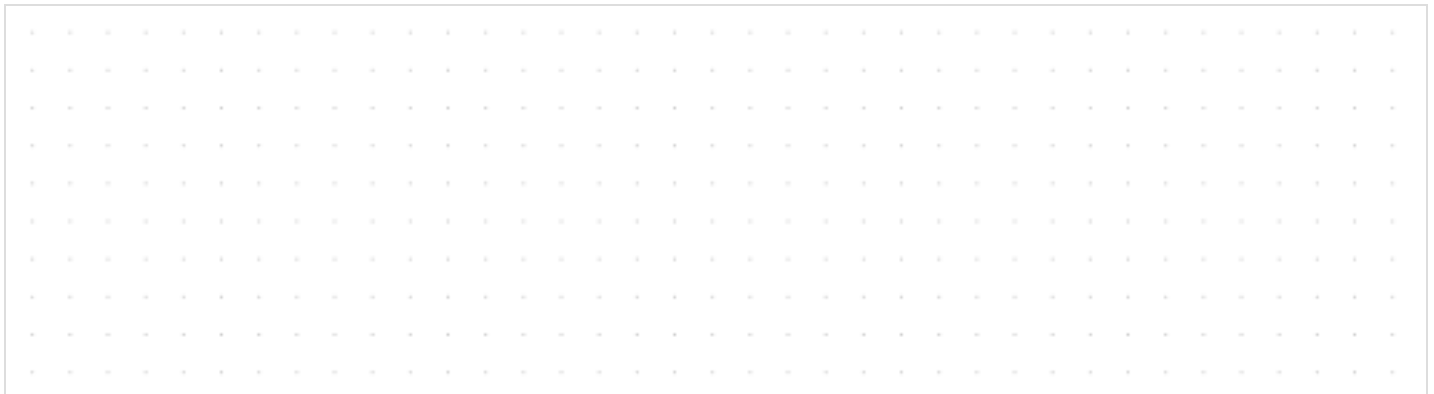


48. A soccer player kicks a soccer ball at an angle. Immediately after the kick, the ball has a horizontal velocity of 20 m/s and a vertical velocity of 12 m/s.

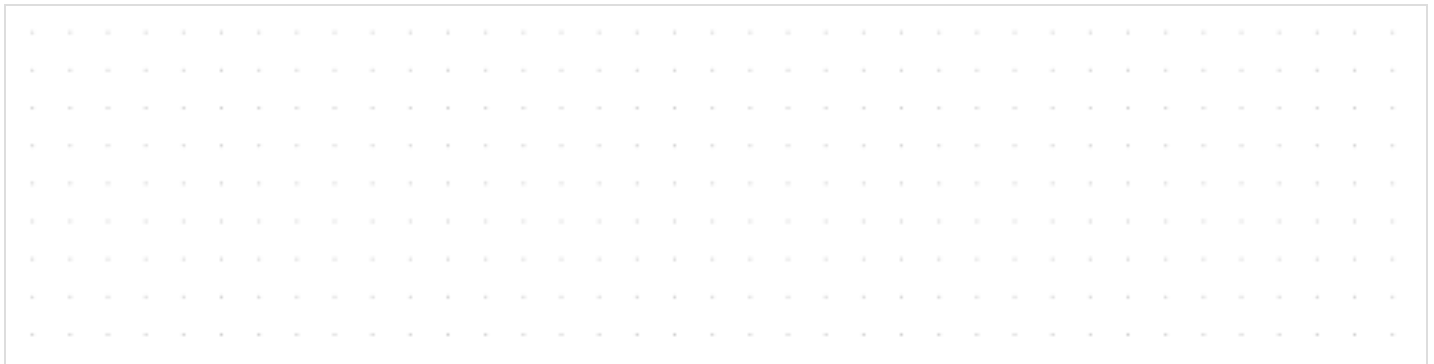
a) How far down the field does the ball fly?



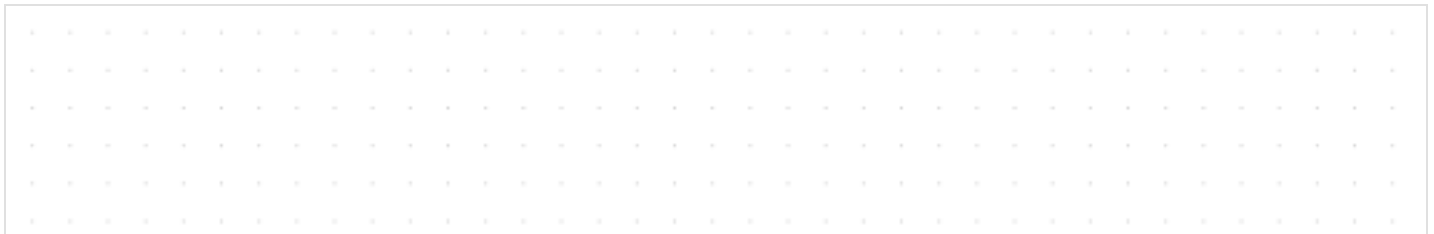
b) With what speed is the ball launched?



c) At what angle does the soccer player kick the ball?



NOTES, WORK & ANNOTATIONS



49. To hit a valid serve in tennis, the ball has to fly over the net, but also land in the service court, which is no more than 6.4 m past the net horizontally on the other side. The net is 12 m away from the baseline where the player hits the ball.

A tennis player hits a serve perfectly horizontally towards the other side of the court with an initial velocity of 40 m/s. The racket makes contact with the ball 2.5 m off the ground, and the height of the tennis net is 0.92 m. **Has the player hit a valid serve?**

Ignore the width of the court and the lateral direction the players move during the game — consider only the dimension along the length of the court and the vertical direction. Ignore air resistance, and therefore any spin-related effects on motion.



“Valid” is **two** separate conditions, and you must check both. Write them down as two inequalities before you compute anything.

NOTES, WORK & ANNOTATIONS

